



RESEARCH ARTICLE

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Key Points:

- We have trained several data-driven models to compress and extract abstract features from solar Extreme ultraviolet full-disk Imaging
- Features extracted using a variational autoencoder are those that correlate best with physical parameters related to the solar activity
- Geomagnetic activity forecasts made from these features are more stable than those made using a baseline pretrained on the ImageNet data set

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Comparison of Solar Imaging Feature Extraction Methods in the Context of Space Weather Prediction With Deep Learning-Based Models

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Abstract Recently, many machine learning-based models have been developed to predict geomagnetic activity several days in advance directly from space-borne solar imaging. To better understand and improve these models, we compare dimensionality reduction techniques to extract abstract features from solar images for space weather-related downstream tasks. We applied the principal component analysis method and trained autoencoders and variational autoencoders (VAE) targeting two reduced dimensions: 1024 and 256, from the original 512×512 images. These methods are compared to a baseline convolutional GoogLeNet model pretrained on the ImageNet data set. We first ensure that the models have been properly trained by evaluating their compression and reconstruction capabilities. We then analyze the latent spaces and find high correlations between the reduced representations of the images and solar activity physical parameters, with the VAE showing the highest potential. In addition, we retrain the SERENADE deep learning model to predict the daily maximum of the K_p index 2 days in advance from the features extracted using each of the dimensionality reduction methods. We find that the predictive accuracy of the new models is similar to that of the baseline model, without any new search for optimized hyperparameters. However, the predictions obtained with the feature extractors dedicated to solar images are much more stable than those obtained with the model pretrained on ImageNet. Furthermore, the analysis indicates that with additional fine-tuning of SERENADE's hyperparameters, the adapted feature extractors could improve predictions during geomagnetic storms, thereby facilitating future advances in days-ahead geomagnetic activity forecasting.

Plain Language Summary Using machine learning models to predict disturbances in the Earth's magnetosphere several days in advance has shown promise, particularly models driven by images of the Sun taken from space. These models use techniques that reduce the large volume of data in solar images to extract the important information contained in them. To improve the performances of these models, we explore and compare several feature extraction techniques like the principal component analysis, autoencoders and variational autoencoders (VAE). We compare these methods to GoogLeNet, a standard reference machine learning model pretrained on an independent image data set (ImageNet). We show that the features extracted with these models have an overall high correlation with physical parameters related to solar activity. When using these reduced representations as inputs to the model SERENADE, which predicts the daily maximum of the K_p index (a measure of the geomagnetic activity), we found that the prediction accuracy is similar to that of the baseline GoogLeNet model. However, the new methods, which were specifically designed for solar images, and especially the VAE, produce more stable predictions. Additionally, with some fine-tuning, the VAE improved predictions during geomagnetic storms, paving the way for improved forecasts of geomagnetic activity several days in advance.

1. Introduction

Space weather as a discipline aims to observe, model, and predict the natural space environment, including near-Earth conditions within the magnetosphere. These conditions are mainly driven by solar activity, which is at the origin of a transfer of energy from the Sun to the Earth's environment, notably via the solar wind and transient structures such as coronal mass ejections (CMEs). These structures can interact with the Earth's magnetosphere and can strongly deform it, therefore causing geomagnetic storms (Gonzalez et al., 1994). These storms have numerous consequences on human activity, mainly disrupting satellite operations, communication systems, and power grids (Horne et al., 2013).

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Over the past few years, the concomitance between the accelerated development of machine learning (ML) methods and the increasing availability of observational data on the various components of the Sun-Earth system (images of the Sun, measurements of interplanetary plasma and magnetic field, measurements of geomagnetic and ionospheric disturbances, etc.) has led to a sustained use of ML-based methods in heliophysics (Asensio Ramos et al., 2023). This is particularly true in the field of space weather, where neural networks and other ML-based methods have been extensively used to nowcast and forecast geomagnetic indices up to a few hours in advance, based on solar wind and other heliophysical parameters, usually measured at the Lagrangian point L1 of the Sun-Earth system (see e.g., references within Camporeale (2019) or Chakraborty and Morley (2020); Tasistro-Hart et al. (2021); Bernoux et al. (2021); Collado-Villaverde et al. (2023); Hu et al. (2023), for more recent examples). These studies have shown that ML-based models are a useful complement to physics-based models, which may lack accuracy or be too computationally expensive, limiting their use in an operational context.

To increase the effective forecasting horizon of geomagnetic indices from a few hours to a few days, recent studies have investigated predicting solar wind parameters at L1, or even the geomagnetic activity, directly from solar imaging, hoping to get a forecast horizon equivalent to the propagation time of the solar wind from the Sun to Earth (i.e., between 2 and 5 days). For instance, Upendran et al. (2020) used a model composed of Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTM) to forecast the solar wind bulk velocity up to 5 days ahead from time-sequences of images of the Sun at 19.3 nm provided by the Atmospheric Imaging Assembly (AIA) instrument onboard the Solar Dynamics Observatory (SDO) spacecraft. Raju and Das (2021) also used the AIA 19.3 nm solar imaging and developed a CNN model to predict the solar wind speed. Similar to Upendran et al. (2020), Bernoux et al. (2022) proposed the model SERENADE, which also relies on stacked CNN-LSTM combination driven by 19.3 nm SDO/AIA images, to forecast the daily maximum of the K_p index (Matzka, Stolle, et al., 2021; Menvielle et al., 2011) 2-to-7 days ahead, another main difference being that their model provides probabilistic forecasts. Hu et al. (2022) use a CNN-based model driven by multiinstrument images from the Solar and Heliospheric Observatory (SOHO) mission to forecast the occurrence of geomagnetic storms as defined by the Dst index (Acero et al., 2018). Unlike the two previous models, the latter has no time dimension in its input and therefore requires no recurrent network. All four studies have shown that these methods, despite some limitations, demonstrate great potential. For instance, SERENADE already outperforms the empirical 27-day recurrence model, which is deemed hard to beat despite its simplicity due to the persistence of solar structures (such as coronal holes) between two apparent rotations of the Sun as seen from Earth (Bernoux et al., 2022; Shprits et al., 2019).

Furthermore, some studies have tried to explore the potential of transformer-type models (Vaswani et al., 2023) as replacements for CNNs. In fact, although CNNs are good in local feature extraction, transformers excel at capturing the global relationships in images. However, transformers need larger data sets for better performance and often require higher computational costs. For instance, E. J. E. Brown et al. (2022) have implemented pre-trained attention-based techniques such as the vision transformer (Dosovitskiy et al., 2021), the transformer in transformer (K. Han et al., 2021), and the swin transformer (Liu et al., 2021) to predict the solar wind speed at L1 using solar imaging at 21.1 nm. When compared to previous works of Upendran et al. (2020), E. J. E. Brown et al. (2022) reported improvements in evaluation metrics such as the root mean squared error (RMSE) and found that their best performing model is the swin transformer. The improved performance observed between the results of E. J. E. Brown et al. (2022) and Upendran et al. (2020) is partially due to the difference in image sampling methods (Asensio Ramos et al., 2023). Specifically, the former employed a 30-min sampling interval, while the latter used daily sampling.

Another possible limitation present in the studies by Upendran et al. (2020), Bernoux et al. (2022), and E. J. E. Brown et al. (2022) lies in their use of pretrained feature extractors. Transfer learning has indeed emerged as a powerful technique in machine learning, particularly when dealing with limited labeled data. By leveraging knowledge from pretrained models on large-scale data sets, transfer learning enables the rapid adaptation of models to new domains (Pan & Yang, 2010). This is invaluable in the context of preliminary studies that require many iterations. It also means more frugal models that can be trained on less powerful machines and consume less energy, making them more sustainable and accessible. However, the training set used for these pretrained models was the ImageNet data set (Deng et al., 2009), made up of millions of images from a thousand categories (planes, boats, cats, dogs, etc.), which have little to do with solar imaging. It would therefore be of great interest to have a pretrained model specialized for solar imagery, capable of providing reduced representations of images that could

be used for various downstream tasks, whether it be predicting the geoeffectiveness of solar activity, the probability of flare occurrence, or other space weather predictions.

In the following study, we tackle the latter problem by developing feature extractors that are specifically adapted and tuned to solar imaging. We do this by comparing different dimensionality reduction techniques. We explore the benefits of using both linear (principal component analysis) as well as nonlinear methods such as autoencoders and compare them to an ImageNet-pretrained baseline. More specifically, we try and address the following two questions: do the extracted features contain physical information about the observed physical properties of interest for space weather applications? Despite the smaller size of solar imaging data sets, are we able to get a tailored feature extractor that performs better than the ImageNet-pretrained baseline?

In the next section, we present the data used in this study, including the preprocessing steps we applied to the solar images. In Section 3, we describe the feature extraction models and methods. In Section 4, we evaluate the ability of these models to reconstruct the original images from the reduced representations. In Section 5, we evaluate the correlation between abstract extracted features of solar images and some physical parameters of solar activity. In Section 6, we train new versions of the SERENADE model to forecast the daily maximum of the K_p index from the features extracted using the different methods and evaluate the obtained forecasts. Section 7 presents additional discussion and conclusions.

2. The Data

In this section, we describe the solar imaging that was used to train the models, together with the preprocessing steps that were applied to the images. We also present the other data that were employed in the analysis of results, including geomagnetic indices and cataloged information about observed solar features.

2.1. Solar Imaging

We use extreme ultraviolet (EUV) images provided by the AIA instrument on board SDO (Pesnell et al., 2012). Launched in 2010, SDO is a mission by NASA for monitoring the Sun. The AIA instrument (Lemen et al., 2012) provides high-resolution and high-frequency images of the Sun in two UV bands, several EUV bands and one visible wavelength. Galvez et al. (2019) published the so-called SDOML data set, which contains ML-ready preprocessed AIA images spanning from 13 May 2010 to 12 December 2020. In particular, in addition to applying all the usual preprocessing routines to obtain Level 1.5 AIA data, that is, images that are despiked, coaligned, flat-fielded, dc-offset, and plate-scale corrected and corrected for exposure time, as explained, for example, in Lemen et al. (2012), the authors removed corrupted images from the data set, reduced its temporal and spatial resolutions (acquisition rate of 6 min, pixel grid of 512×512 with a pixel size of ≈ 4.8 arcsec), and reencapsulated the data set in a Zarr container (Miles et al., 2020).

Four AIA channels were selected for this study: 17.1, 19.3, 21.1, and 30.4 nm, each associated with a different spectral and temperature response (Lemen et al., 2012). These wavelengths were selected for their observational characteristics, as together they provide a global overview of the quiet and the hotter region corona. For instance, the 19.3-nm channel, for which the primary emission lines are the Fe XII (peaking at $\sim 1.5 \times 10^6$ K) and the Fe XXIV (peaking at $\sim 20 \times 10^6$ K), shows the active regions, coronal holes, and the solar flares, which are of interest to our subsequent goal, that is, geomagnetic activity prediction. The rest of the channels offer additional observational advantages to the 19.3-nm one by allowing more insights onto the quiet times of the Sun's corona (17.1 nm), active regions (21.1 nm) and the transition regions (30.4 nm). For more details about the characteristics and temperature response of these channels, we refer the reader to Lemen et al. (2012) and Boerner et al. (2012).

2.2. Additional PreProcessing Steps

We adapt the preprocessing steps used in Bernoux et al. (2022) for each wavelength used in our analysis. To enhance specific characteristics of the images, such as pixel contrast, we apply a clipping procedure where the values of the images are restricted between specific minimum and maximum thresholds. These thresholds were determined through an empirical testing, building upon the results of Ahmadzadeh et al. (2019). After iterative testing, the optimal values that we found are 10 data numbers per second (DN/s) as a minimum and 4000 DN/s as a maximum for the wavelength 17.1 nm, 20 DN/s and 5000 DN/s for 19.3 nm, 10 DN/s and 9000 DN/s for 21.1 nm, and 1 DN/s and 300 DN/s for the wavelength 30.4 nm (see Figure 1 for an example on a 19.3 nm image).

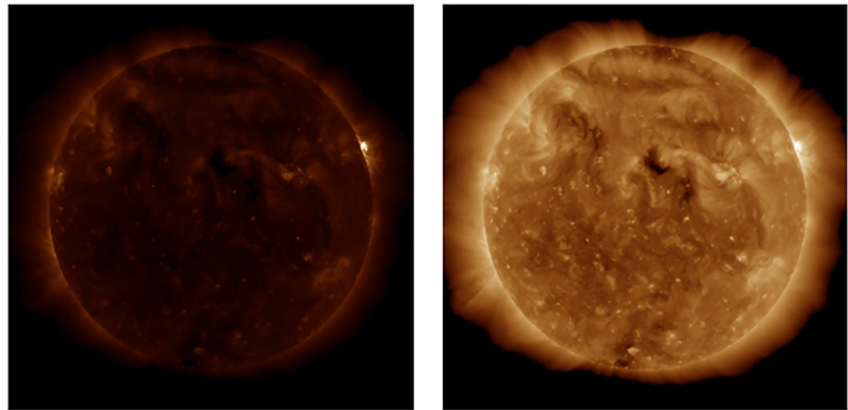


Figure 1. Image of the Sun at the wavelength 19.3 nm on 9 September 2010 12:00 by SDO/AIA. On the left hand side is the image from the SDOML data set. The right hand side shows the same image after clipping and logarithmic scaling.

Following this adjustment for each wavelength, we further normalize the images by rescaling them to a $\log_{10}$ scale to mitigate data distribution skewness. These steps ensure that the data have good numerical properties for learning, prevent the models from being biased by varying scales, and improve its overall performance.

Finally, we apply transformations to the images that serve as data augmentation techniques during training only, artificially increasing the size of the data set. We apply a horizontal flip, followed by a vertical flip each with a probability of 50% chance of being applied. In total, that gives us four possible cases of 25% probability each to transform the data: both flips are applied; only the horizontal flip is applied; only the vertical flip is applied; and no transformation is applied returning the original image. Certainly, artificially increasing the training data set does not generate “real” data points from a scientific perspective, but the transformed images are considered as variants of the original ones, showing different points of views of the same content. By applying data augmentation techniques, we improve the model's robustness and enhance its generalization performance, especially when dealing with a limited data set.

2.3. Data Separation

The subsets used for training, validation, and testing the models should satisfy 3 conditions: being independent from one another, being large enough to ensure as much examples as possible during the different training steps, and to be as representative as possible of the global variation of the data set. In our case, that means that each subset must contain periods that correspond to the different phases of the solar cycle (minimum, ascending, maximum, and descending). It is worth noting that the data set period spans the 24th solar cycle, characterized by notably low levels of solar activity, posing challenges in satisfying the 3 aforementioned conditions for data separation.

We separate our data set into yearly folds with the following partition (see Figure 2):

1. Training data set: 2010, 2012, 2013, 2015, 2016, 2018, and 2019;
2. Validation data set: 2011;
3. Test data set: 2014, 2017, and 2020.

In summary, we use a data set made up of images at four EUV wavelengths: 17.1, 19.3, 21.1, and 30.4 nm from the SDOML data set. We are primarily interested in the 19.3 nm channel. However, we observed that the feature extraction models performed better when provided with additional data. To increase the number of training samples, we incorporated the three additional channels as a data augmentation technique. It results therefore in a data set with 2,175,180 training samples and 301,201 validation samples. As for the testing subset, we keep 244,441 samples solely of images from the 19.3 nm channel. The images are 6 min apart and are of a 512×512 pixel resolution.

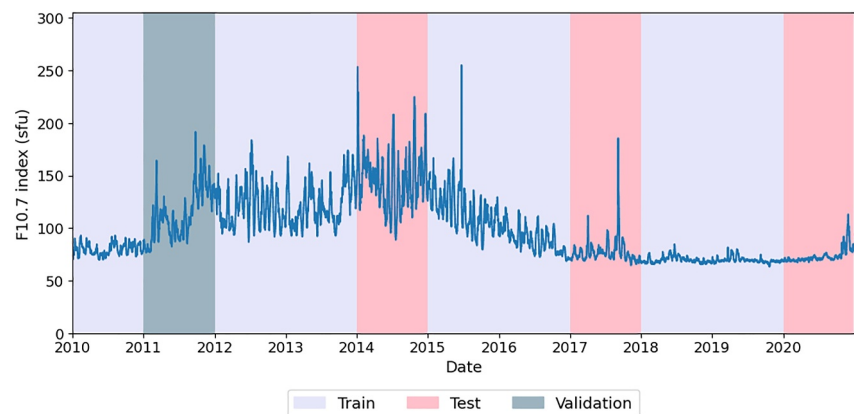


Figure 2. Partitioning of the data set into training, validation, and test subsets. The F10.7 index is displayed to give an indication of solar activity and serve as a proxy for the solar cycle.

2.4. Indices and Catalogs

We use observations that characterize the solar and geomagnetic activities to assess the performance of the models and evaluate the interpretability of the reduced dimensions. We consider data that describe the solar cycle and structures such as coronal holes and active regions.

We use the solar radio flux at 10.7 cm, that is, the F10.7 index as a solar activity and cycle indicator (Tapping, 2013). It was first measured in the year 1947 and is currently provided by the National Research Council Canada. It is a robust index with few gaps and issues as it can be monitored on a daily basis and is not affected by the weather. It originates high in the solar chromosphere and low in the corona, making it valuable to track important regions.

We describe the solar characteristics using the properties of active regions and coronal holes. Active regions (ARs) are bright areas of intense and complex magnetic field. They are typically associated to solar flares and CMEs (Toriumi & Wang, 2019). These regions are characterized by high-intensity emissions in the UV and X-ray wavelengths. Their intensities vary along the solar cycle and they appear brighter than the surrounding plasma on the solar disk. On the other hand, coronal holes (CHs) are regions characterized by open magnetic field lines that allow the acceleration of the escaping solar wind into the interplanetary space. They appear darker in the UV and X-ray wavelengths due to their low plasma density and temperature. The coronal holes are more frequent and persistent during solar minimum periods and can last through several of the Sun's rotation period (27 days). During high solar activity, we observe these structures at midlatitudes and along the solar equator. Both the intensities of active regions and the areas occupied by coronal holes are crucial in understanding solar dynamics and for space weather predictions. We retrieve the intensity of observed active regions and the surface occupied by coronal holes from the year 2010 until 2021 from the Heliophysics Events Knowledgebase (HEK, Hurlburt et al., 2012) repository. The properties were determined using the recognition method spatial possibilistic clustering algorithm (SPoCA, Barra et al., 2008; Barra et al., 2009). The method uses a multichannel, unsupervised fuzzy clustering algorithm (Bezdek, 2013) to automatically divide EUV solar images into coronal holes, quiet sun, and active regions. The method was successfully applied to data from different instruments such as SOHO/EIT, STEREO/EUVI, PROBA2/SWAP, and SDO/AIA (Verbeeck et al., 2014). It was integrated in the HEK for near-real time cataloging of AR and CH from SDO/AIA. These physical properties will be used as targets in the search for information in the reduced spaces.

Furthermore, we use data from the GOES series spacecrafts that carry the X-ray Sensor instrument (XRS, Chamberlin et al., 2009) that monitor the X-rays coming into the Earth's upper atmosphere and provide alerts about X-ray flares. We use the X-ray flux data set that provides measurements in different band passes (0.05–0.4 nm or short channel and 0.1–0.8 nm or long channel). We use the 1-min average products of the 1-s cadence data in the long channel as it is more sensitive to solar flares and is used for flare classification.

Finally, in order to assess the geomagnetic perturbations, we use the geomagnetic index Kp . It stands for the Planetary K-index and it is used to indicate disturbances in the Earth's magnetic field (Matzka, Stolle, et al., 2021).

Furthermore, many physics-based models are driven by the K_p index. Those include magnetospheric and plasmaspheric models (Bourdarie & Maget, 2012; Glauert et al., 2014; Subbotin & Shprits, 2009; Tu et al., 2013) as well as ionospheric models (Bruinsma et al., 2012). The K_p index has been measured since 1932 using 13 observatories spread on the globe (11 northern and 2 southern stations). A K_p value equal or greater than five indicates a geomagnetic storm.

3. Feature Extractors

In this section, we introduce the feature extractors applied to the solar imaging in this study: GoogLeNet, principal component analysis, autoencoders, and variational autoencoders. For all the models, we describe their structures and architectures and give details about their training. The models were trained using the designated training set, which included data from the years 2010, 2012, 2013, 2015, 2016, 2018, and 2019.

3.1. GoogLeNet

The GoogLeNet (GN, Szegedy et al., 2015) model was originally developed for image classification purposes. It is a deep CNN formed by 9 Inception modules on top of each other. One Inception module is made of several convolution layers with different kernel sizes, 1×1 , 3×3 , and 5×5 set in parallel, which allow the model to extract the information in at different scales from the input data. The modules are then followed by a single dense layer and finally by a softmax activation layer that serves as the classifier. In order to use the GN model as a dimensionality reduction technique, the upper layers specialized in classification were removed. The model was trained on the large-scale ImageNet data set (Deng et al., 2009) of labeled images which has become a benchmark for training and validating computer vision algorithms. The GN model takes as input RGB-like images resized to 224×224 pixels. The multichannels are obtained by copying the single channel 3 times for each input solar image individually. The GN method reduces the input data to a 1024-dimensional representation (hereafter referred to as GN-1024). We also explore the advantages of a more compact representation of 256 components. For that, we use a principal component analysis transformation (see below) on the GN's 1024 components and keep only the 256 first components (hereafter GN-256). The GN model was used as a feature extractor in Bernoux et al. (2022) and Upendran et al. (2020) and will be considered as the baseline model.

3.2. Principal Component Analysis

Principal component analysis (PCA) is possibly the most used algorithm in multivariate statistics (Greenacre et al., 2022). The goal of this dimensionality reduction method is to find an orthogonal basis on which we can efficiently project or express the data. In a nutshell, PCA decomposes the input data as a weighted sum of orthogonal linear combinations of the original data. The principal components (or basis vectors) can be found by calculating the eigenvectors of the covariance matrix of the observations. The eigenvalue of each vector represents the amount of variance that this vector carries. The vectors are then put in a descending order according to their eigenvalue. Once the basis is constructed and the number of components to keep is chosen, the reduced data set can be obtained by projecting the original data on the directions, afterward one can reconstruct the original data by multiplying each projection by the basis. When applying this method on the solar imagery, we analyze the results using both the 1024 components (hereafter PCA-1024) from each image and the 256 components (PCA-256). Figure 3 shows the cumulative variance explained by each of the 1024 components when the PCA is applied to the 4 wavelengths. We show that with this number of components, we can retrieve up to 95% of the total variance in the data set and 92% with 256 components.

3.3. AutoEncoders and Variational AutoEncoders

An autoencoder (AE, Hinton & Salakhutdinov, 2006) is a design of an artificial neural network intended to learn an efficient, more compact representation of data, often for dimensionality reduction and feature extraction purposes. An AE consists of two main blocks: an encoder and a decoder. The encoder compresses the input data (in our case, an image) into a lower dimensionality latent space (LS), capturing the most important features. The decoder then reconstructs the original data from this compressed representation. The model is trained by minimizing the error between the input and the reconstructed output using a cost function.

A variational autoencoder (VAE, Kingma & Welling, 2022) is an extension of the standard AE that introduces a probabilistic approach to the latent space generation, making it useful for generative tasks. Unlike the standard

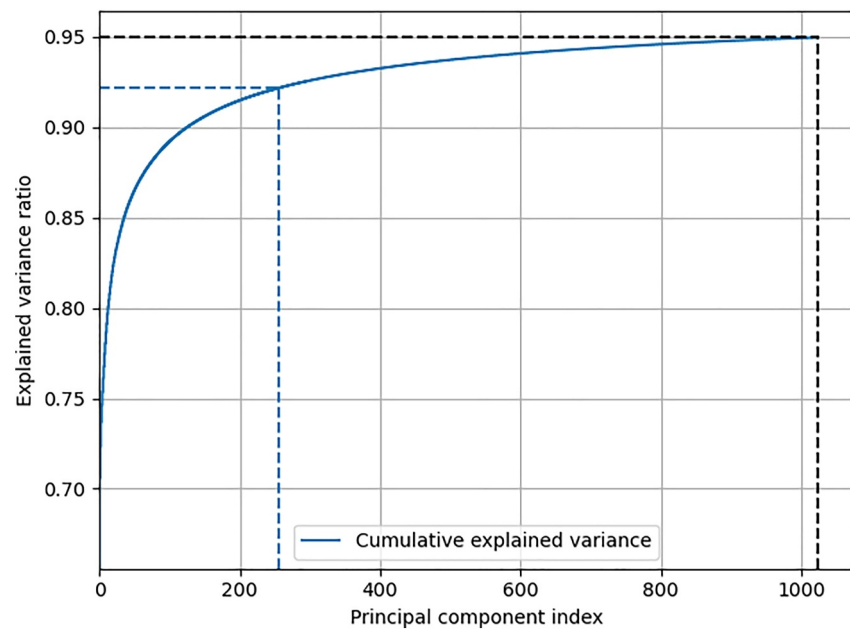


Figure 3. Cumulative explained variance of each of the 1024 PCA components applied on the 4-wavelength data set. The total explained variance of the 1024 components is equal to 95% and 92% for the 256 components, as indicated by the dashed lines.

AEs, which map the input in a deterministic way to the latent space, VAEs model the latent space as a distribution, typically a Gaussian, from which samples can be drawn. The encoder is therefore modified to output parameters defining a distribution (a mean and a variance in the case of a Gaussian distribution) instead of a single point, and then the latent representation is sampled from it. The decoder then reconstructs the input data from the sampled latent representations. The VAE is trained by minimizing a loss function that includes both the reconstruction loss and a regularization term (the Kullback-Leibler divergence) that controls the learned distribution to be as close as possible to a prior, typically a standard normal distribution. A β -VAE introduces an adjustable hyperparameter β into the VAE's loss function that controls the trade-off between the reconstruction accuracy and the disentanglement of the latent space.

AEs, along with their variants, excel in various domains, including computer vision tasks (Geng et al., 2015; Hou et al., 2020), image denoising (Tripathi, 2021), distinguishing contours and objects in the images (Redmon et al., 2016; Zhang et al., 2019), data generation (C. Han & Wang, 2021; Huang & Jafari, 2023; Tian et al., 2023), anomaly detection (Ding et al., 2022; Ying et al., 2023), and text processing (Abdi & Williams, 2010; García-Mendoza et al., 2022). AEs are proving to be useful in the field of solar physics and heliosphysics.

For instance, E. Brown et al. (2021) used a sigma-VAE (Rybkin et al., 2021) on images from SDO AIA and Helioseismic and Magnetic Imager (HMI, Scherrer et al., 2012) instruments. The images include the 9 EUV channels from AIA and 3 HMI channels measuring the strength of the components of the magnetic field. The authors evaluate the performance of the VAE's latent space to predict the F30 index (a proxy for solar UV irradiance, de Wit & Bruinsma, 2017) and compared the results with those of the ResNet feature extractor (an ImageNet pretrained model He et al., 2016) and the persistence model. The VAE outperformed the ResNet model notably for prediction horizons of 1 and 2 days but showed no significant out-performance compared to the persistence model. J. Wang et al. (2022) trained an autoencoder on the Space-Weather HMI Active Region Patch (SHARP) data to detect anomalies prior to a strong flare. The aim is to determine indicators of strong solar flares to improve their prediction and alert time. The model successfully predicted a large number of strong flare events of classes M and above and had an average warning time of 45.24 hr. Giger and Csillaghy (2024) used a method that combines a VAE and a context encoder (ce-VAE) developed for medical imaging (Zimmerer et al., 2019) to heliophysics purposes. Their goal was to detect anomalies or uncommon patterns in the images in order to retrieve interesting features related to the solar activity. The authors applied this method on the SDOML data set using the 17.1-nm wavelength with a spectral resolution of 256×256 in order to reduce the images to a latent

$$\text{MSE}(I_{\text{orig}}, I_{\text{recon}}) = \frac{1}{n \times n} \sum_{i,j=1}^n (I_{\text{orig}}(i,j) - I_{\text{recon}}(i,j))^2 \quad (1)$$

It is worth noting that while the MSE is a simple and easily implemented metric, it is poorly correlated with the human perception of visual systems. It is also boundless which makes it hard to interpret in our case.

The SSIM, on the other hand, is a metric that aims to mimic the human perceptual system by measuring the differences between images on three components: the luminance $l(x,y)$, contrast $c(x,y)$, and structure $s(x,y)$. The latter are defined as

$$l(x,y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}, \quad (2)$$

$$c(x,y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}, \quad (3)$$

$$s(x,y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}, \quad (4)$$

where x and y are two spatial patches from each image, μ_x , μ_y , σ_x^2 , σ_y^2 , and σ_{xy} are the mean of x and y , the variance of x and y , and the covariance of x and y , and finally C_1 , C_2 and C_3 are constants included to avoid instability and stabilize the divisions (Z. Wang et al., 2003).

The values of the SSIM are scaled between -1 and 1 . The higher the value, the greater the similarity between the compared images. The value is given by the weighted combination of the three components:

$$\text{SSIM}(I_{\text{orig}}, I_{\text{recon}}) = l(I_{\text{orig}}, I_{\text{recon}})^\alpha \cdot c(I_{\text{orig}}, I_{\text{recon}})^\beta \cdot s(I_{\text{orig}}, I_{\text{recon}})^\gamma \quad (5)$$

where $\alpha = \beta = \gamma = 1$.

Figure 5 shows examples of reconstructions obtained with the reduced representation of the same image taken on 9 September 2014. Visually, we see that the images are recognizable in all of the reconstructions and that the PCA reconstructions are crispier than those obtained with the AEs and the VAE. This is expected as the latter methods reconstruct the images via neural networks (decoders) that lose some information during the process, contrary to the PCA that reconstructs through a linear projection with less losses. To quantify the comparison, we show the metrics' results in Table 2. For the MSE metric, the models AE-1024 and PCA-1024 show very similar values with the PCA-1024, slightly outperforming the AE-1024. As for the AE-256 and PCA-256 models, the MSE differences are minimal to none, while the VAE-256 performs slightly poorer than the two models. The SSIM metric results favor the AE-1024 model to the PCA-1024. Similarly, the AE-256 and VAE-256 have similar scores and outperform the PCA-256. It is expected to see a slight decrease in the reconstruction performance between the AE and the VAE, as the latter includes a regularization term in its training that shifts the “focus” from the reconstructions to the organization of the latent space. However, the fact that the differences are small indicates a good trade-off between the two aspects.

In a nutshell, we see a minimal deviation in the MSE and SSIM results between the different models. The lower MSE for the PCA models indicates that the original and reconstructed images are numerically close, while the AE and VAE's higher SSIM values indicate that the reconstructed images maintain a similar structure and perceptual quality to the original images. Generally, the results show that the models were correctly fitted and trained, as they manage to compress and decompress the images with small errors.

5. Reduced Space Analysis

In this section, we aim to identify information within the reduced representations that describe both the solar cycle and geomagnetic activity. To do this, we search for correlation between the reduced representations and the targeted observations. The evaluation was done on the same testing set presented in Section 2.3 (on the years

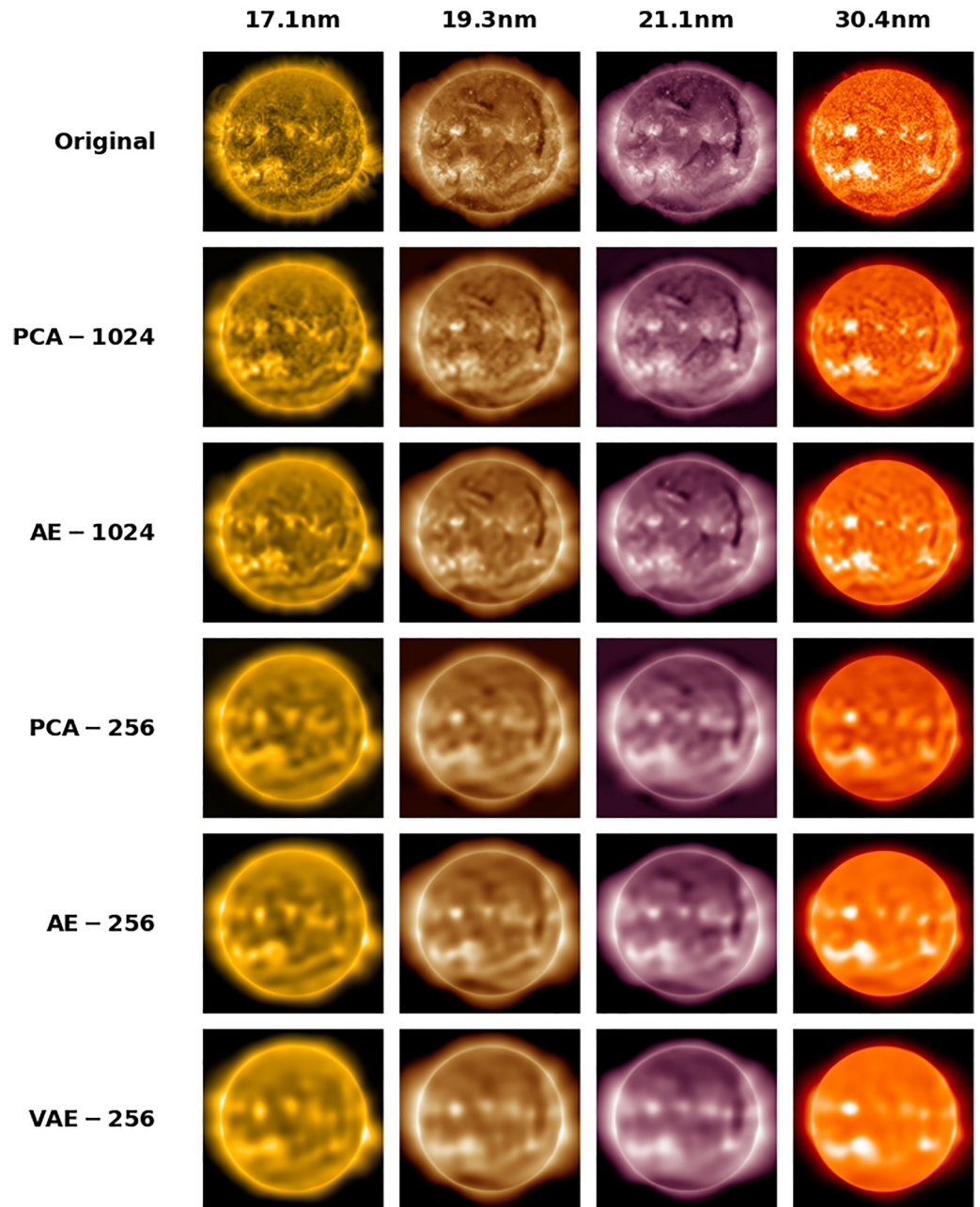


Figure 5. Reconstructions of an image taken on 2014-09-09. From left to right, we plot the same image at different wavelengths: 17.1, 19.3, 21.1, and 30.4 nm. From top to bottom: original image, image reconstructed using the PCA-1024, AE-1024, PCA-256, AE-256, and VAE-256.

2014, 2017, and 2020). The results discussed in this section are derived solely from the embeddings of the 19.3 nm images. This wavelength was chosen due to its observational advantage, as it clearly highlights the features of interest in our study (Lemen et al., 2012).

We aim to understand to which degree the components of the reduced representations are related to the observed properties of the solar cycle. Therefore, we perform a linear regression to fit the reduced representations to the observations of those properties. These include the F10.7 index, the maximum intensity of observed ARs, the total area occupied by CHs, and the X-ray flux observed by the GOES satellites for each wavelength separately. For each case, we consider the reduced representations and data from the testing years, 2014, 2017, and 2020, to fit the

Table 2

Values of the Metrics for the Evaluation of the Quality of the Reconstruction of Solar EUV Images From Their Reduced Representation

Metric	Model	17.1 nm	19.3 nm	21.1 nm	30.4 nm
MSE	PCA-1024	1.4E − 3	0.9E − 3	0.7E − 3	1.8E − 3
	AE-1024	1.7E − 3	1.1E − 3	1.0E − 3	2.0E − 3
	PCA-256	2.6E − 3	1.9E − 3	1.5E − 3	2.3E − 3
	AE-256	2.5E − 3	1.9E − 3	1.6E − 3	2.4E − 3
	VAE-256	2.8E − 3	2.3E − 3	2.0E − 3	2.6E − 3
SSIM	PCA-1024	0.72	0.81	0.83	0.61
	AE-1024	0.78	0.88	0.91	0.64
	PCA-256	0.65	0.76	0.77	0.60
	AE-256	0.74	0.85	0.90	0.62
	VAE-256	0.74	0.84	0.88	0.62

Note. Models are grouped by their output sizes (1024 and 256). The best value for each EUV channel per model group is shown in bold.

linear model. Next, we calculate the correlation between the predictions of the linear fit and the observations over those 3 years.

In Table 3, we show the correlation percentages obtained for the 19.3-nm wavelength between the linear fit predictions and the observations. Our analysis focuses on this wavelength because it is a versatile and informative one, allowing us to observe both coronal hole features and the Sun's eruptive activity. Additionally, this is the wavelength used to train the SERENADE prediction model (Bernoux et al., 2022).

The 1024-dimensional models generally perform better than the 256 ones as they scored higher correlations values. For the F10.7 index and the AR intensity, the three 1024-dimensional models perform similarly with correlation of 99% and 84%, respectively. As for the CH area and the X-ray flux, the PCA-1024 and the AE-1024 slightly outperform the GN-1024 model. The 256-dimensional models also show high correlation percentages with the 4 considered observations. It can be seen that the performances of the PCA-256 and the AE-256 are very close because they yield similar correlation percentages, both outperforming the GN-256. The VAE-256 model, on the other hand, stands out a little more than the rest of the models as it has scored the

highest correlation percentages with the observations, specifically excelling in the AR intensity and the CH area cases. The improvement of the correlation results between the AE-256 and VAE-256 shows the importance of a regularized latent space while training these types of dimensionality reduction techniques. It is also worth noting that although the GN model is pretrained on an independent database, it has however produced good results compared to adapted feature extractors. Additionally, since the GN-256 method (to recall, this technique was obtained by applying a PCA on the GN-1024 extracted features) shows a significant decrease in performance compared to the GN-1024 model, we decide to omit the GN-256 from the rest of the evaluation and keep the models where we apply the PCA directly on the solar imaging (PCA-1024 and PCA-256).

6. Using the Extracted Features to Forecast the Geomagnetic Activity

We now evaluate the impact of using the features extracted from the images by the different methods on the prediction of the daily maximum of the geomagnetic index Kp (denoted $Kp_{\max,24hr}$) several days in advance. To do this, we use the SERENADE model, a geomagnetic activity prediction ML-based model driven directly and solely by solar imaging. We first describe the SERENADE model, then train several instances of it, driven by the different feature extractors, and evaluate the resulting forecasts on several aspects: regression, classification, and robustness.

Table 3

Correlation Percentages Obtained With the Predictions of the Linear Fit and the Observed Properties: the F10.7 Index, the AR Intensity, the CH Area, and the X-Ray Flux

Model	F10.7	AR intensity (DN/s)	CH area (km ²)	X − ray flux (W/m ²)
GN-1024	99%	84%	92%	80%
PCA-1024	99%	84%	94%	85%
AE-1024	99%	84%	95%	85%
GN-256	96%	78%	83%	69%
PCA-256	96%	78%	86%	76%
AE-256	96%	76%	89%	75%
VAE-256	97%	81%	95%	79%

Note. The AR intensity unit DN/s stands for data number per second, which represents the raw digital output of the instrument's detector, corresponding to the number of photons detected and converted to an electrical signal. Models are grouped by their output sizes (1024 and 256). The best value for each parameter per model group is shown in bold.

6.1. The SERENADE Model

SERENADE is a neural network composed of three sequential modules: a feature extractor, a temporal encoder, and a decoder. In this section, we review the original SERENADE model developed by Bernoux et al. (2022).

The feature extractor takes as input the transformed images at the wavelength 19.3 nm, and as was mentioned previously, the authors opted here for transfer learning techniques and used the GN-1024 model in order to reduce computational time and cost. A sequence of 10 images is given to the feature extractor over a period of 5 days with a time step of 12 hr between two consecutive images. If the sequence contains a missing image at a certain time, we use the image that is closest to it with a maximum spread of 6 hr. If this time is exceeded, the sequence is disregarded. Therefore, the sequence of 10 images is transformed by the feature extractor to a tensor of dimension 10×1024 that summarizes the useful spatial information contained in the images.

The temporal encoder is a neural network that extracts information from the temporal aspect of the data. SERENADE uses RNNs of the LSTM type to learn long-term dependencies. It is composed of two layers of 2048 hidden units. The module generates a 2048-dimensional vector of spatiotemporal features.

Finally, SERENADE's decoder (the multilayer perceptron or MLP) transforms the spatiotemporal information into forecasts of $Kp_{\max, 24hr}$. To generate probabilistic predictions, which are more useful from an operational point of view, SERENADE outputs two values: the mean and the standard deviation, which define a normal (Gaussian) distribution. For this purpose, the decoder is composed of two parallel branches, each containing three dense layers. The size of the layers decreases gradually: the first layer takes as input the 2048 features and has an output dimension of 512, the second 64, and finally the last layer outputs a single value. To train the model, the authors used a combination of RMSE and Gaussian negative log likelihood loss functions. Several instances of SERENADE were trained for forecast horizons ranging from 2 to 7 days.

In this study, we use and compare the five new feature extractors that we have developed in place of the original GN-1024 feature extractor. As in the previous section, we use the extracted features of the 19.3-nm channel for evaluating the forecasts. We do so for a more consistent comparison with the works of Bernoux et al. (2022) as it is the one that the authors used in their study. For each feature extractor, we will train 10 instances of SERENADE by varying the random seed (in order to add randomness to the initial conditions and therefore estimate the metrics' uncertainties), but we will keep all the training hyperparameters used by Bernoux et al. (2022). This means that the model structure is optimized for use with the GN-1024 model instead of the other models, which will undoubtedly benefit the performance of this model in particular, and to some extent the feature extractors that output 1024-dimensional vectors instead of 256-dimensional ones. We make this choice to keep this study concise and focused on its initial problem.

We train all our models with a forecast horizon of 2 days. We use the same data set split used to train the feature extractors (see Section 2.3).

6.2. Forecast Evaluation Methodology

We assess the model's performance from several angles. First, we focus on the regression aspect, using metrics such as root mean squared error (RMSE) and continuous ranked probability score (CRPS). The RMSE measures the difference between two series and puts more emphasis on larger errors. It is given by

$$\text{RMSE}(X, Y) = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2}, \quad (6)$$

where $X = (x_i)_{1 \leq i \leq N}$ and $Y = (y_i)_{1 \leq i \leq N}$ are time series representing here the model forecast and the true observations. N is the number of samples. The CRPS is a measure of the deviation of a scalar value from a probability distribution. It can be considered as a probabilistic generalization to the mean absolute error (MAE). It is given by

$$\text{CRPS}(\mathcal{F}, Y) = \frac{1}{N} \sum_{i=1}^N \int_{\mathbb{R}} (\mathcal{F}_i(x) - \mathbf{1}(x \geq y_i))^2 dx, \quad (7)$$

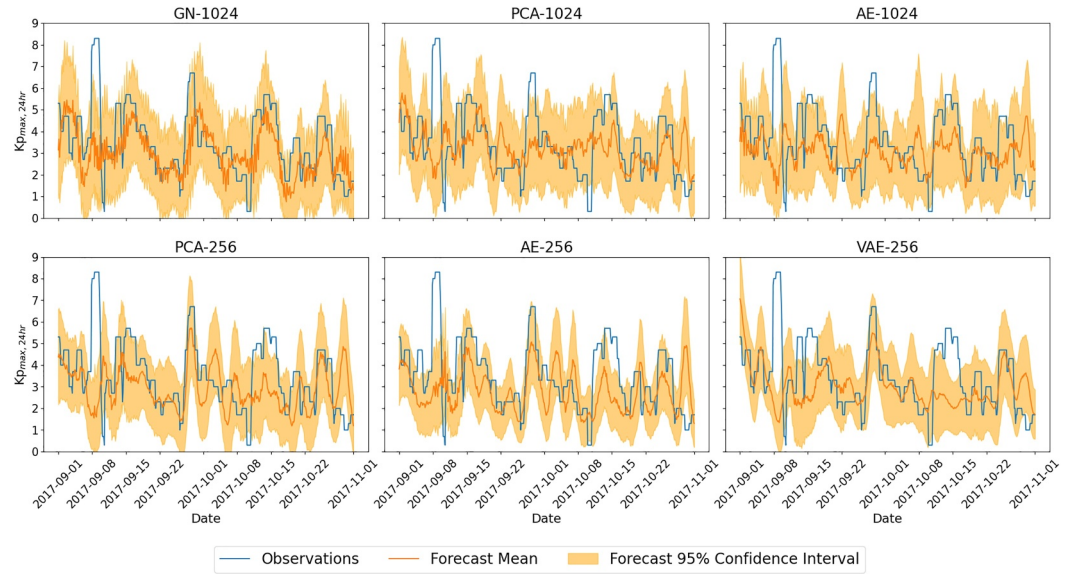


Figure 6. Comparing the original SERENADE model predictions (GN-1024) with the newly developed feature extracting methods. The first row shows the predictions using (from left to right) GN-1024, PCA-1024, and AE-1024. The second row shows results using PCA-256, AE-256, and VAE-256. The blue line indicates the true values of $Kp_{\max,24hr}$, the orange lines and areas represent the forecast mean and 95% confidence interval.

where $\mathcal{F} = (\mathcal{F}_i)_{1 \leq i \leq N}$ represents the series of probability densities associated with each probabilistic forecast. Lower RMSE and CRPS values are better.

To get a more complete picture, we calculate these two metrics over four subsets. The first one is the “full” set and comprises the entire test period (the years 2014, 2017, and 2020). The second one are the “calm periods,” where we only keep the quiet periods ($Kp_{\max,24hr} < 4^-$). The third and fourth subsets are active periods, where we distinguish between CME-associated storms (denoted as “CME storms”) and SIR-associated ones (labeled as “SIR storms”). To distinguish between these two types of storms, we filter the data using catalogs obtained from Nguyen et al. (2025) for CMEs and for SIRs.

We also assess the classification performance of the model by analyzing its ability to distinguish quiet from disturbed periods. For that, we use the Brier score (BS) that allows us to evaluate probabilistic predictions of binary classes. It is defined as

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i - C_i)^2, \quad (8)$$

where $p_i \in [0, 1]$ is the predicted probability of being in a disturbed period and $C_i \in \{0, 1\}$ is the true observed class (0 denoting quiet periods and 1 denoting disturbed periods). Here, we define disturbed periods as $Kp_{\max,24hr} \geq 4^-$. The BS metric is analogous to the mean squared error (MSE) between the predicted probability being in a certain class and its corresponding observation. Lower BS values are better.

6.3. Forecast Results Analysis

Figure 6 shows examples of predictions obtained with SERENADE over the period from 1 September 2017 to 1 November 2017 for each of the feature extractors. Since we trained 10 instances of SERENADE for each feature extractor, we display here the results from the models that obtained the best CRPS values on the full test set. We can immediately see that the results obtained differ significantly in terms of precision, stability, and dynamics, without it always being obvious how to rank them. That is why we also analyze these results using the measures defined above. We provide the regression results in Figure 7.

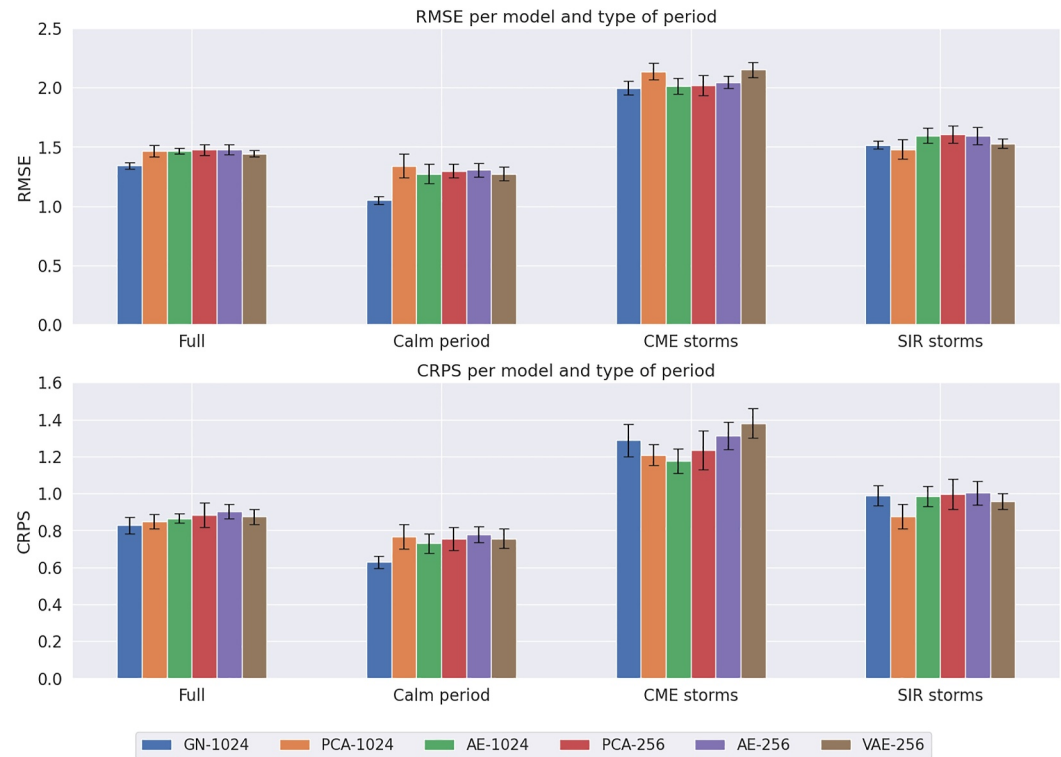


Figure 7. Mean and standard deviations of the (a) RMSE and (b) CRPS over 10 trainings of the SERENADE model per feature extractor per perturbation period over 10 randomly initialized models. The error bars indicate a 1σ deviation from the mean.

For the “full” and “calm” periods, we see similar behavior for both metrics, where the GN method has the lowest values with an average of 1.34 for the RMSE and 0.87 for the CRPS in the first case and 1.05 for the RMSE and 0.63 for the CRPS in the second case. The rest of the models show comparable performances in these cases. For “CME storms”, the GN-1024 method scored an average RMSE of 2.0, similar to the AE-1024, PCA-256, and AE-256. The models PCA-1024 and VAE-256 show slightly worse performance with an RMSE average of 2.15 (7.2% difference). As for the CRPS results during CME storms, the GN model was outperformed by the PCA-1024, AE-1024, and PCA-256, scoring an average of 1.28, 1.20, 1.18, and 1.23, respectively. The AE-256 model matched the performance of the GN, while the VAE-256 has the highest CRPS value of 1.38 (7.5% increase). For SIR storms, the GN-1024, PCA-1024, and the VAE-256 methods have the lowest average RMSE (1.52 and 1.48, respectively), while the rest of the models perform very similarly (1.59). For the CRPS results, the GN-1024 model (CRPS = 1.0) was outperformed by the PCA-1024 and then the VAE-256 with CRPS values of 0.88 and 0.96.

The classification evaluation results are shown in Figure 8, which provides the BS for all models and test years individually. In the year 2014, the 1024 dimensional models outperform the 256 dimensional ones with little overlapping in the error bars: The AE-1024 scored the best value, while the VAE-256 scored the highest error value. In 2017, the GN-1024 outperforms the rest of the models but shows however a large error range, while the VAE model has a narrower deviation. In the year 2020, the BS score of all the models drops significantly compared to the two previous years. This year is in a solar minimum dominated by quieter periods. These results are therefore consistent with what we found in Figure 7 for the “calm period” subset that showed the least error between the different conditions. The best performing models in this year are the GN-1024 and the VAE-256. Additionally, all models exhibit rather low BS scores (BS < 0.3) and that suggests that their probabilistic forecasts are reliable.

In addition, we evaluate the credibility and stability of the predictions. Indeed, a noticeable aspect in the predictions of Figure 6 is the tendency of the GN-1024 method to produce unstable, highly oscillating predictions compared to the other methods, which can hinder operational use. To determine whether the new models provide

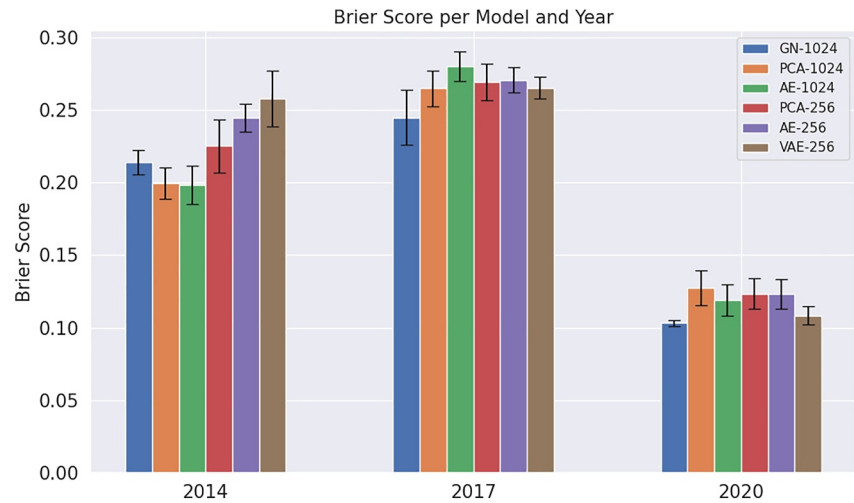


Figure 8. Mean of the Brier score results over 10 trainings of the SERENADE model per feature extractor per test year. The error bars indicate the standard deviation.

more credible forecasts, we first compute and compare the time-lagged autocorrelations for the observations and the forecasts produced by each model. Figure 9 shows these results for time lags ranging from 0 to 60 days. The autocorrelation of the forecasts was computed using the three test years. The autocorrelation of the observations was computed over the full range of the SDOML data set (11 years), and a confidence interval is estimated by iteratively recomputing it over all 3-year combinations. The autocorrelation of the true $Kp_{\max,24hr}$ values follows the expected behavior, that is, a rapid degradation of autocorrelation over the first 3 days, then a peak in autocorrelation (around 0.3) 27 days later, and a second smaller peak at 54 days. The autocorrelation of the GN-1024 forecasts is much higher, with a magnitude about twice as large, and systematically outside the confidence interval, although both peaks are observed. The consistent overestimation of the strength of the autocorrelation may indicate a form of overfitting of the model and a lack of generalization properties. The other five models demonstrate superior performance in capturing the temporal correlation of the output data, with autocorrelation values falling within the established confidence interval. The VAE-256 model seems to reproduce the autocorrelation curve particularly well, with both peaks clearly reproduced with the right amplitude.

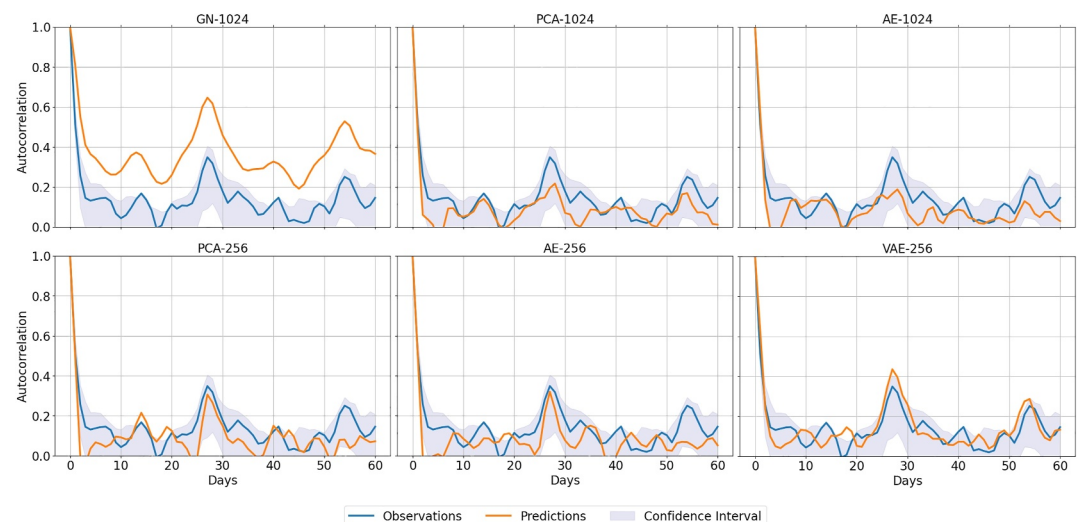


Figure 9. Comparison between the time-lagged autocorrelation of the $Kp_{\max,24hr}$ values and the autocorrelation of the forecasts provided by each model. The autocorrelation of the predictions is calculated over the three test years. The autocorrelation of the observations is calculated over the period 2010–2020, as well as over all combinations of three years within this period, to provide a confidence interval.

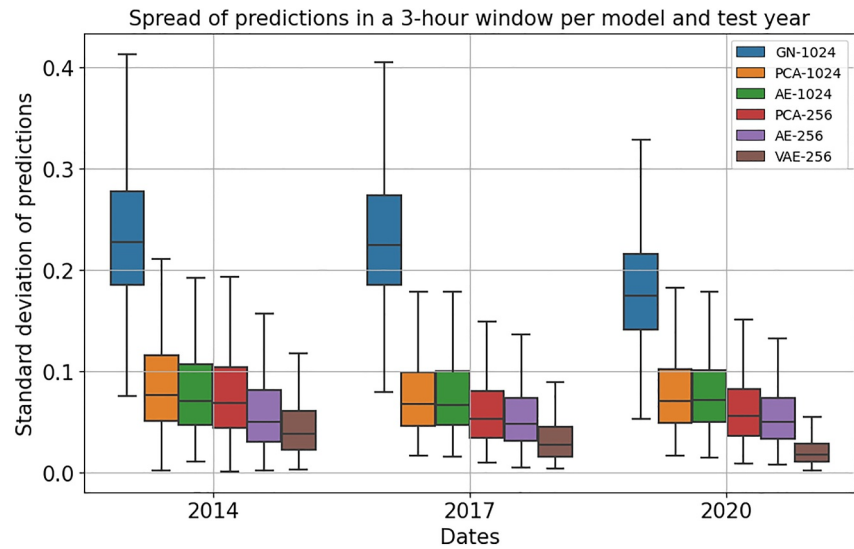


Figure 10. Box plot of the distribution of standard deviations of the forecasts obtained within each 3-hr window in which $Kp_{\max,24hr}$ values do not change. Lower standard deviations denote a higher model stability. Results are grouped by the feature extractor and test year.

Furthermore, based on the premise that Kp , and thus $Kp_{\max,24hr}$, is a 3-hourly index (i.e., values change only every 3 hr), we assume that predictions should remain fairly consistent within these 3-hr windows. To evaluate the stability of the models, we generate $Kp_{\max,24hr}$ forecasts with SERENADE every 6 min and measure the spread of the forecasts over all the 3-hr windows. Figure 10 shows a box plot summarizing the distribution of the standard deviation of forecast values within all 3-hr windows for each model and test year individually. We observe the same behavior for each of the test years, where the GN-1024 model has the highest standard deviations between forecasts within the same windows (as well as the largest “spread of forecast spread”). Thus, the median value of the spread of forecasts obtained with model GN-1024 in 2014 is 0.22, with a maximum of 0.41, which is the order of magnitude of the interval between two Kp values (remember that this is a discrete index with increments around 0.3). Conversely, the other models show much lower values, with the VAE-256 model showing the smallest prediction deviations, with a maximum of 0.12 also reached in 2014. Feature extractors trained on solar images are therefore the source of much more stable and robust forecasts, which is a highly desirable property for a forecaster.

6.4. Effect of SERENADE's Architecture on the Predictions

The results in Section 6.3 show that while the GN-1024 features provide the SERENADE model with the highest accuracy, the statistical study shows that the VAE-256 features generate more stable predictions. Knowing that the SERENADE hyperparameters used previously were optimized by Bernoux et al. (2022) for use with GoogleNet, in this section, we investigate whether optimizing the hyperparameters for use with VAE-256 could improve the forecasts obtained with it. After iterative search and testing over the hyperparameter space, we propose an alternative architecture for SERENADE. One of the main changes was to the layer sizes of the LSTM and MLP modules. The LSTM is now composed of one layer of 256 hidden units instead of two layers of 2048 hidden units, thus dividing the number of LSTM parameters by a factor of 16. The decoder model is still composed of two branches, but the dense layers are now of sizes 64, 32, 16, and 1 instead of 512, 64, and 1. As a result, the new optimized architecture is much more compact, which makes sense since each image is now described by 256 features instead of 1024. All details about these modifications can be found in the Appendix Appendix B.

As in the previous sections, we train 10 instances of the new SERENADE model by varying the random seed in order to estimate the uncertainties of the results. In Figure 11, we show a comparison between the results obtained with the new model (hereafter VAE-256-new) and the old model (VAE-256-old) as well as the baseline GN-1024. We show an example of the predicted time series over the same period as Figure 6 (panel a) along with the

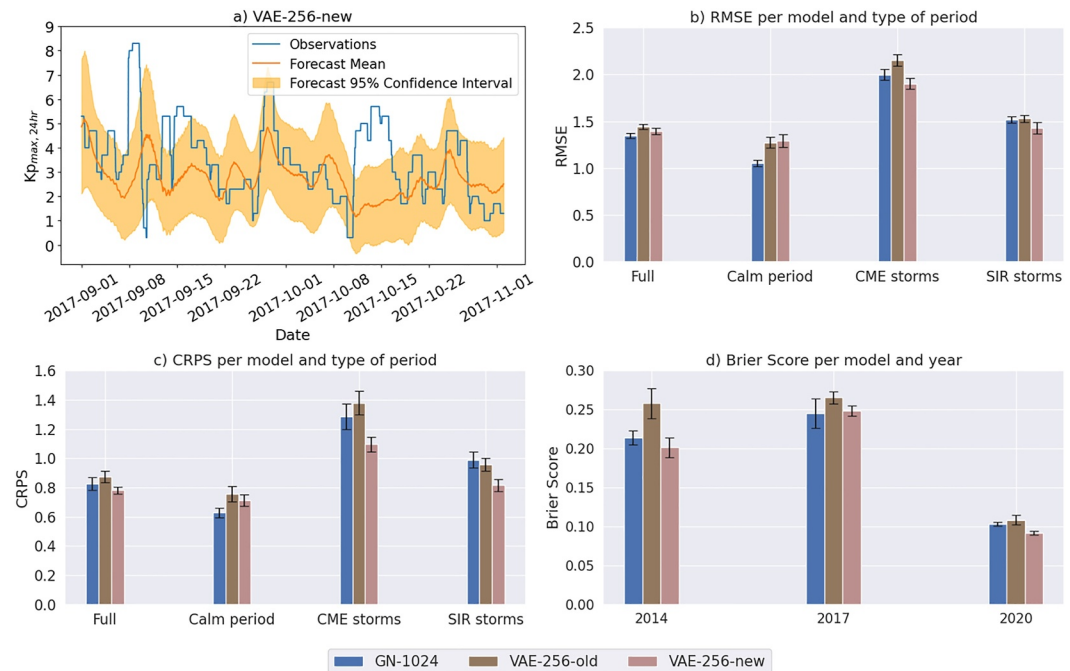


Figure 11. Analysis of the forecast results obtained with a VAE-256-optimized SERENADE model (VAE-256-new). (a) $Kp_{max,24hr}$ prediction example using the VAE-256-new SERENADE model. (b) RMSE comparison per model (GN-1024, VAE-256-old, and VAE-256-new) and type of period. (c) CRPS comparison per model and type of period. (d) Brier score comparison per model and testing year (2014, 2017, and 2020). The average of the metrics and the error bars were obtained after 10 different training iterations of the SERENADE model.

regression (RMSE, in panel b, and CRPS, in panel c) and classification (BS, in panel d) metrics similar to Figures 7 and 8. We first notice a significant improvement in the RMSE and CRPS values, especially during the storm periods. For instance, in the case of CME storms, the mean value of the RMSE metric decreased from 2.15 to 1.90 (a 12.3% improvement), while the CRPS decreased from 1.38 to 1.09 (a 23.5% improvement). Furthermore, the BS for the year 2014 (a period of solar maximum) decreased by 26.1%. During disturbed periods, the new model is even much more accurate than the GN-1024 model, particularly in terms of RMSE, but even more so in terms of CRPS. This suggests that the model was better able to learn to estimate forecast uncertainty from the latent characteristics of VAE-256 than from GN-1024. However, during calm periods, the GN-1024 model is still more accurate.

In short, the modifications made to the SERENADE architecture have significantly improved the performance of the VAE-256-driven model, making it the most efficient during periods of geomagnetic disturbance, while maintaining its high stability. This indicates that it is possible to improve the predictive performance of models driven by feature extractors adapted to solar images and thus to better exploit the information contained in their latent spaces. This opens many avenues for future improvement of the SERENADE model, including deeper modifications in its architecture and training process.

7. Discussion and Conclusions

We have presented and compared several approaches to develop solar imaging-tailored feature extraction methods for use in the context of space weather prediction. We tested three methods: principal component analysis, autoencoders, and variational autoencoders, which we compared to a GoogLeNet model pretrained on the ImageNet database. We have shown that both PCA and AE are better at reconstructing original images from compressed latent representations than VAE, which sacrifices reconstruction quality in favor of a better organized latent space. This allows the VAE to extract features that, for the same vector size, appear to contain more information about solar activity than features extracted by the other methods. Latent spaces of dimension 1024 offer greater correlations than those of dimension 256, but the difference seems small given the compression ratio

between the two. It is worth noting that the GN method also showed a rather high correlation with solar features, even though it was trained on an independent data set.

However, it is the GN model that produces the features that, when used in the downstream task of predicting $Kp_{\max,24hr}$, provide the SERENADE model (using the same hyperparameters as in Bernoux et al., 2022) with the best values of the metrics that assess model accuracy. This seems to suggest that having been trained on a much more diverse database (1,281,167 training images divided into 1000 different classes) is an advantage. Although it is known that CNNs learn local patterns and generic abstract features that can be reused for other tasks (Zeiler & Fergus, 2014), it is still somewhat counterintuitive to imagine that a model that has not been pretrained on solar images would produce the best predictions for this space weather task. We can also hypothesize that this is because the GN model, having been trained to classify images, has learned to produce features that are more easily exploitable for a downstream task than latent features. Yet, this observation should be tempered by the fact that the other models based on PCA, AE, and VAE perform almost as well.

Moreover, these models, and especially the model based on the VAE, produce prediction results for $Kp_{\max,24hr}$ with dynamics that are much more credible and stable than the GN model. This is a very important point. Indeed, it is hard to imagine that a forecaster would be satisfied with a model whose forecasts vary as much within a few dozen minutes as is the case with the GN model. This also seems to suggest to us that the models dedicated to solar images have indeed learned better certain aspects related to the global physics of the system. From this point of view, and considering that the VAE produces the small latent space that correlates best with the physical properties, it seems that the latter model developed here has the most potential of the three. This is why we aimed to quickly optimize SERENADE's pipeline hyperparameters for use with the VAE, without modifying its architecture. We found that these minor modifications significantly improved SERENADE's predictions, especially for storms associated with CME and SIR, which opens many promising avenues. It is also worth noting that the PCA and the pretrained GN give good results and are very easy and quick to implement, opening up the possibility of developing prototype models based on them fairly quickly.

It is also possible to test dual pretraining, using a VAE pretrained successively on an external data set, such as ImageNet, or the new data set The Well, which provides images of physical systems and simulations (Ohana et al., 2024) and on solar images. It would also undoubtedly be advantageous for such pretraining to use images not only from SDO/AIA but also from other missions, most notably SOHO/EIT (Delaboudinière et al., 1995). This would extend the temporal coverage to a much more active solar cycle (from 1995 onward) and thus increase and diversify the training data set. We therefore believe that an effort to obtain an ML-ready data set from SOHO/EIT that is properly cross-calibrated and interoperable with an SDO/AIA ML-ready data set (whether SDOML or another) would be a great asset to the community. Other avenues for improvement lie in the learning techniques used. It would be interesting to check the results obtained with models that have learned with contrastive losses (Le-Khac et al., 2020), or with true multimodal learning that would combine images from different instruments (magnetograms, coronograms, etc.) and their representations, tending toward a foundation model for solar and coronal physics.

Appendix A: Additional Implementation Details—Feature Extractors

This appendix gives additional implementation details, such as the loss function, optimization technique, and learning rate used in the training of the feature extractors.

A1. Principal Component Analysis

For applying this method, we use the IncrementalPCA from Python's Scikit-Learn decomposition package (Pedregosa et al., 2011) iterating over the batches of the data set. We fit one PCA on 1024 components and keep the first 256 when needed.

A2. AutoEncoders

Optimization method: We train the autoencoder with the backpropagation method. We use the Adam optimization method (Kingma & Ba, 2017) with a learning rate $lr = 2E - 5$ and a weight decay of $1E - 6$. We optimize the training with the ReduceOnPlateau scheduler with a patience of 20 epochs and a reduction factor of 0.75.

Loss function: We use the MSE loss function. In order to help the autoencoder reconstruct the images accurately and therefore extract the physical information in its latent space, we create a multiterm loss function. The total loss is given by

$$MSE_{total} = \alpha MSE(I_{orig}, I_{recon}) + \beta MSE(d_{x_{orig}}, d_{x_{recon}}) + \gamma MSE(d_{y_{orig}}, d_{y_{recon}}) \quad (A1)$$

where I_{orig} and I_{recon} denote the original and reconstructed images, respectively, and d_x and d_y denote the horizontal and vertical gradients of the images. We add the terms α, β and γ as weights, where $\alpha = 1, \beta = \gamma = 2$. The purpose of incorporating these terms is to add more definition to the features' contours and edges, resulting in a clearer reconstruction.

Epochs, batch size: Each autoencoder (256 and 1024) is trained on a maximum of 100 epochs with a batch size of 32. We load the data using 4 workers.

Dropout: We insert dropout layers between the CNN blocks with a rate of 0.3.

Software and hardware: The autoencoder was developed and written using the libraries PyTorch (v.2.0.0, Paszke et al., 2019) and PyTorch Lightning (v. 1.8.6, Falcon & The PyTorch Lightning team, 2019) for Python (v. 3.10.3). We also used the packages Numpy (v. 1.26.4, Harris et al., 2020), Pandas (v. 2.2.2, McKinney, 2010), Matplotlib (v. 3.8.4, Hunter, 2007), Seaborn (v. 0.12.2, Waskom, 2021), and Zarr (v. 2.17.2, Miles et al., 2020).

Precision: We encode our data as half-precision floats (16 bits instead of 32) to reduce the learning time.

Total training time: We trained both autoencoders for a total duration of 2 weeks (14 days) on a computing server with one Nvidia RTX6000 GPU.

A3. Variational AutoEncoder

Optimization method: We train the variational autoencoder with the backpropagation method. We use the Adam optimization (Kingma & Ba, 2017) method with a learning rate $lr = 1E - 5$ and a weight decay of $1E - 6$. We optimize the training with the ReduceOnPlateau scheduler with a patience of 20 epochs and a reduction factor of 0.75.

Loss function: We use a two-term loss function, one for the reconstruction and another for the regularization term: The first one is identical to the one used for the autoencoders. The regularization term loss is the Kullback-Leibler term. The total loss is expressed as

$$MSE_{total} = \alpha MSE + \beta KL_{div} \quad (A2)$$

where $\alpha = 1$ and $\beta = 10E - 2$.

Epochs, batch size: The VAE-256 is trained on a maximum of 100 epochs with a batch size of 32. We load the data using 4 workers.

Dropout: We insert dropout layers between the CNN blocks with a rate of 0.3.

Software and hardware: The VAE was developed and written using the libraries PyTorch (v.2.0.0, Paszke et al., 2019) and PyTorch Lightning (v. 1.8.6, Falcon & The PyTorch Lightning team, 2019) for Python (v. 3.10.3). We also used the packages Numpy (v. 1.26.4, Harris et al., 2020), Pandas (v. 2.2.2, McKinney, 2010), Matplotlib (v. 3.8.4, Hunter, 2007), Seaborn (v. 0.12.2, Waskom, 2021), and Zarr (v. 2.17.2, Miles et al., 2020).

Precision: We encode our data as half-precision floats (16 bits instead of 32) in order to reduce the learning time.

Total training time: We trained the variational autoencoder for a total duration of 2 weeks (14 days) on a computing server with one Nvidia RTX6000 GPU.

Appendix B: Modified SERENADE Model (VAE-256-New)

This appendix gives additional implementation details about the modified architecture of the model SERENADE to adapt it to the newly developed feature extractors VAE-256.

Optimization method: We use the AdamW optimization (Loshchilov & Hutter, 2019) method with a learning rate $lr = 5E - 5$ and a weight decay of $5E - 4$. We optimize training with the MultiStepLR optimizer with a reduction factor of 0.5 every 10 epochs.

Loss function: We use a combination of RMSE and Gaussian negative log likelihood loss functions (See Bernoux et al., 2022).

Epochs, batch size: The model was trained on a maximum of 10 epochs with a batch size of 32. We load the data using 4 workers.

Dropout: We insert dropout layers between the LSTM layers and the decoder branches of rate 0.3.

Precision: We encode our data as half-precision floats (16 bits instead of 32) to reduce the learning time.

Data loading: The data were loaded with sequences of 10 images over a period of 12 hr with a sequence stride of 1 hr.

Total training time: We trained the SERENADE models for a total duration of 33 min on a computing server with one Nvidia RTX6000 GPU.

Software and hardware: The SERENADE model was trained using the libraries PyTorch (v.2.0.0, Paszke et al., 2019) and PyTorch Lightning (v. 1.8.6, Falcon & The PyTorch Lightning team, 2019) for Python (v. 3.10.3). We also used the packages Scikit-Learn (v. 1.4.2 Pedregosa et al., 2011), Numpy (v. 1.26.4, Harris et al., 2020), Pandas (v. 2.2.2, McKinney, 2010), Matplotlib (v. 3.8.4, Hunter, 2007), and Properscoring (v.0.1, Corporation, 2024).

Data Availability Statement

The K_p index data are available from GFZ Data Services (Matzka, Bronkalla, et al., 2021) with CC BY 4.0 license. The SDOMLv2 data set is available from NASA via an AWS S3 Bucket (Meng & Wright, 2019) with no restrictions on the use of this data. The pretrained GoogLeNet was built using the default implementation provided by the torchvision package (TorchVision maintainers and contributors, 2016). The model weights are the same as the original paper (Szegedy et al., 2015).

The F10.7 index data are available from the OMNIweb data services (Papitashvili & King, 2020). The active regions intensity and coronal holes surface data are available from the Heliophysics Events Knowledgebase with Creative Commons Attribution Noncommercial License (Hurlburt et al., 2012). The GOES X-ray flux data are available from the NOAA National Centers for Environmental Information.

The ICME and SIR event catalogs can be downloaded from Nguyen et al. (2025) with etalab 2.0 license.

All models used in this study were built using publicly available Python packages. All details and references needed to reproduce the models are given in the paper and in the appendices. We make the data needed to reproduce the results and figures, that is, the extracted features for the entire period covered by the SDOML data set for the 19.3 nm channel and the SERENADE outputs for the test years (with the varied seed and the 6 min forecasts), the instances and weights of the AEs and VAE publicly available online (Tahtouh et al., 2025) with etalab 2.0 license.

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